

Functional Requirements Document (FRD)

System name: SCD2_DLT_Pipeline

1. Purpose and Scope

- Purpose: Define functional specifications to ingest CSV data into Delta, cleanse, transform, build SCD Type 2 ordersummary, compute customer aggregate spend, and orchestrate via Delta Live Tables (DLT).
- In scope: CSV ingestion (customer, order), schema application, TotalAmount derivation, null-like and duplicate removal, SCD Type 2 ordersummary, customeraggregatespend, DLT orchestration.
- Out of scope: Security, SLAs, environments, monitoring, downstream BI, unspecified entities/columns/systems.

2. Actors and Systems

- Actors: Data Engineer (), Platform Scheduler (), DLT runtime.
- Systems: Volume storage (CSV), Delta Lake tables, Delta Live Tables.
- Environments: .

3. Data Sources and Targets

- Sources: customer CSV at /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata; order CSV at /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata.
- Targets: Delta tables customer, order (catalog/schema); ordersummary at gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary; customeraggregatespend at gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend.

4. Data Schemas

- customer: CustId, Name, EmailId, Region (types).
- order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId (types).
- ordersummary: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date, RecordStatus, StartDate, EndDate (types).
- customeraggregatespend: Name, TotalAmount, Date (types).

5. Functional Requirements

- FR-CMN-001 Title: Ingest CSV to Delta (customer, order)
- Description: Read CSVs from volumes and load into Delta tables customer and order.
- Preconditions: Source CSVs available; target storage accessible.
- Main Flow:

1) Read customer CSV from configured path.

2) Read order CSV from configured path.

3) Write datasets to Delta tables customer and order.

- Alternate Flows:
- A1: Source not found -> mark pipeline failed; log error; stop.
- Acceptance Criteria:
- AC1: customer and order Delta tables exist and contain rows when sources present.
- AC2: Fail with clear error when sources missing.
- FR-SCH-002 Title: Apply explicit schemas
- Description: Enforce defined schemas for customer and order on read or before write.
- Preconditions: FR-CMN-001 executed or source streams available.
- Main Flow:

1) Define schema for customer fields.

2) Define schema for order fields.

3) Apply schemas to ingested data.

- Alternate Flows:
- A1: Type mismatch -> cast where safe; else quarantine row ().
- Acceptance Criteria:
- AC1: Columns and order match definitions.
- AC2: Rows violating schema are handled per A1 or .
- FR-TRN-003 Title: Derive TotalAmount in order
- Description: Add TotalAmount = PricePerUnit * Qty.
- Preconditions: FR-SCH-002 complete for order.
- Main Flow:

1) Compute TotalAmount expression.

2) Persist updated order table/view for downstream steps.

- Alternate Flows:
- A1: Null PricePerUnit or Qty -> TotalAmount null.
- Acceptance Criteria:
- AC1: TotalAmount equals product for valid numeric inputs.
- AC2: Column available for downstream join/aggregation.
- FR-DQ-004 Title: Remove null-like values and duplicates

- Description: Clean customer and order by removing records containing "Null"/"null" per business context and dropping duplicates.
- Preconditions: FR-TRN-003 completed for order; FR-SCH-002 applied.
- Main Flow:

1) Identify rows with any field containing "Null" or "null" string.

2) Exclude identified rows.

3) Deduplicate remaining rows (all columns as key).

- Alternate Flows:
- A1: If dedup key needs refinement -> use all columns by default.
- Acceptance Criteria:
- AC1: No rows remain where any field equals "Null"/"null" as string.
- AC2: No duplicate rows remain.

- FR-TGT-005 Title: Create ordersummary if not exists
- Description: Ensure ordersummary table exists in gen_ai_poc_databrickscoe.sdmc_wizard.
- Preconditions: Catalog/schema accessible.
- Main Flow:

1) Check for table existence.

2) Create with defined schema including SCD metadata if absent.

- Alternate Flows:
- A1: Missing catalog/schema -> fail with error.
- Acceptance Criteria:
- AC1: ordersummary present with required columns.

- FR-JOIN-006 Title: Join customer and order on CustId
- Description: Produce joined dataset for SCD Type 2 loading.
- Preconditions: FR-DQ-004 complete; FR-TGT-005 complete.
- Main Flow:

1) Inner join customer and order on CustId ().

2) Select target columns for ordersummary base.

- Alternate Flows:
- A1: If join type must include non-matching -> .
- Acceptance Criteria:

- AC1: Joined rows contain fields per ordersummary non-SCD columns.
- FR-SCD-007 Title: Implement SCD Type 2 load for ordersummary
- Description: Maintain history with Active/Inactive, StartDate, EndDate on natural key CustId, OrderId.
- Preconditions: FR-JOIN-006 complete; FR-TGT-005 complete.
- Main Flow:
 - 1) Identify current active records by key.
 - 2) Detect changes across any tracked attributes (Name, EmailId, Region, ItemName, PricePerUnit, Qty, Date).
 - 3) For changed keys, set prior record EndDate to current load timestamp; set RecordStatus to Inactive.
 - 4) Insert new record with StartDate=current load timestamp; EndDate=null; RecordStatus=Active.
 - 5) For new keys, insert Active record with StartDate set; EndDate=null.
- Alternate Flows:
 - A1: Late-arriving data handling -> treat as new change with StartDate at load time ().
- Acceptance Criteria:
 - AC1: Only one Active record per key at any time.
 - AC2: Complete historical chain with non-overlapping date ranges.
 - AC3: Status values exactly Active/Inactive.
- FR-SCH-008 Title: Ensure ordersummary schema
- Description: Validate presence of required business and SCD columns.
- Preconditions: FR-SCD-007 executed at least once.
- Main Flow:
 - 1) Verify required columns exist.
 - 2) Add missing columns if permissible ().
- Alternate Flows:
 - A1: If immutable schema -> fail with descriptive error.
- Acceptance Criteria:
 - AC1: ordersummary contains declared columns with correct names and case.
- FR-TGT-009 Title: Create customeraggregatespend if not exists
- Description: Ensure target table exists in gen_ai_poc_databrickscoe.sdlc_wizard.
- Preconditions: Catalog/schema accessible.

- Main Flow:

1) Check for table existence.

2) Create with columns Name, TotalAmount, Date.

- Alternate Flows:

- A1: Missing catalog/schema -> fail.

- Acceptance Criteria:

- AC1: customeraggregatespend present with required columns.

- FR-AGG-010 Title: Aggregate spend by Name and Date

- Description: Compute SUM(TotalAmount) from ordersummary grouped by Name and Date.

- Preconditions: FR-SCD-007 complete; FR-TGT-009 complete.

- Main Flow:

1) Select Active records from ordersummary.

2) Group by Name, Date.

3) Compute SUM(TotalAmount) as TotalAmount.

4) Load results into customeraggregatespend (upsert/overwrite).

- Alternate Flows:

- A1: No Active records -> write empty partition/set.

- Acceptance Criteria:

- AC1: Values equal sum of contributing orders.

- AC2: No duplicate Name-Date rows after load.

- FR-ORC-011 Title: Orchestrate pipeline using Delta Live Tables

- Description: Implement steps FR-CMN-001 through FR-AGG-010 in DLT with deterministic DAG.

- Preconditions: DLT pipeline configured; permissions granted.

- Main Flow:

1) Define DLT tables/views with dependencies matching DAG.

2) Configure execution order via DLT dependencies.

3) Execute pipeline; monitor status.

- Alternate Flows:

- A1: Step failure -> DLT marks table failed; halt downstream; surface error.

- Acceptance Criteria:

- AC1: DLT run produces all target tables populated per preceding FRs.
- AC2: DAG reflects Appendix K order.
- AC3: Re-runs are idempotent for unchanged inputs.

6. Business Rules Alignment

- BR-TotalAmount: PricePerUnit * Qty enforced in FR-TRN-003.
- BR-DataQuality: Remove "Null"/"null" and duplicates enforced in FR-DQ-004.
- BR-SCD2: CustId+OrderId key; Active/Inactive; StartDate/EndDate enforced in FR-SCD-007.
- BR-Aggregation: Group by Name, Date; SUM(TotalAmount) enforced in FR-AGG-010.

7. Assumptions and Constraints

- Assumptions: Missing data types and join type are .
- Constraints: Use only specified columns/expressions; preserve names/case; DLT used for orchestration.

8. Error Handling and Logging

- Strategy: Fail-fast on missing sources/targets; log descriptive errors in DLT event logs; retry policy .

9. Security and Access

- Access controls, PII handling, and compliance: .

10. Performance and SLA

- Volume, latency, schedule, and SLAs: .

11. Test Scenarios and Acceptance (traceable to FRs)

- T1 -> FR-CMN-001: customer and order populated.
- T2 -> FR-TRN-003: TotalAmount correct.
- T3 -> FR-DQ-004: No "Null"/"null" or duplicates.
- T4 -> FR-JOIN-006: Joined records present.
- T5 -> FR-SCD-007: SCD2 fields and history correct.
- T6 -> FR-AGG-010: Aggregates correct by Name, Date.
- T7 -> FR-ORC-011: DLT orchestrates end-to-end.

12. Open Items

- Data types for all columns: .
- Join type for FR-JOIN-006: (default inner).
- Scheduling/triggers for DLT: .

- Quarantine/reject handling for bad rows: .
- Upsert/overwrite strategy for aggregates: .

Mermaid diagrams

System flow diagram

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Process flow / flowchart

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

Data flow diagram (DFD)

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

Sequence diagram

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High-level architecture diagram

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